

Chapter 63 — Paediatric Urologic Oncology: Renal and Adrenal

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NEUROBLASTOMA

- **Commonest extracranial solid tumour of childhood; over half present with metastases.**
- Arises from **neural crest** cells of the adrenal medulla and sympathetic ganglia — anywhere along the **sympathetic chain** (neck, thorax, retroperitoneum, pelvis, adrenal).
- **75% retroperitoneal — 50% adrenal, 25% paravertebral.**
- Behaviour ranges from **spontaneous regression** through **differentiation to a benign tumour** to **extreme malignancy**; **biological factors** explain much of this variance.

Epidemiology and Genetics

Incidence

- **8–10% of childhood cancers; 10 per million live births** (United States); **commonest malignancy of infancy.**
- **Median age 19 months**; most under 5 years; **almost all diagnosed by 10 years.**

Genetics

- Familial cases (<1%): **autosomal dominant; median age 9 months; ≥20% bilateral adrenal or multifocal** (rare in sporadic disease); sibling or offspring risk <6%.
- Linkage to **16p12–13**; predisposition genes **PHOX2B** and **ALK**.

Constitutional Chromosome Abnormalities

- Deletions, translocations and amplification all carry **prognostic weight** (frequencies and direction in the marker table under Biologic Variables).
- **Aneuploidy** here means an abnormal chromosome number (hyperdiploidy); it is favourable against **diploid or tetraploid** content.
- **MYCN amplification = more than 10 copies** at 2p24; drives **rapid progression and treatment failure**. Neuroblastoma was the **first tumour whose chemotherapy intensity was set by biological markers**, not stage and histology alone.
- **1p deletion**: consensus region **1p36.1–2**, candidate suppressor **CHD5**; independent; constitutional 1p abnormalities also reported.
- **Unbalanced 11q loss of heterozygosity (LOH)** = LOH of 11q **with retention of 11p** (versus whole-chromosome 11 LOH). 1p loss travels with MYCN amplification; **11q loss rarely does**, and flags **poor outcome in low-stage, non-amplified** tumours.
- **17q gain** (often by translocation with 1 or 11; 17q22–qter): on multivariate analysis the **strongest prognostic factor**, then **stage IV**, then **1p deletion**.

Embryology and Spontaneous Regression

- **In situ neuroblastoma** (term of Beckwith and Perrin): incidental adrenal neuroblast nodules, histologically identical to neuroblastoma, found at autopsy in infants under 3 months at **40–45 times the clinical incidence** → most **regress**. Such nodules exist in **all fetuses**.

- **Prenatally detected** tumours follow a **favourable course**; neonatally resected adrenal masses, **cystic or solid**, mostly carry **favourable markers**.
- Expectant observation of screen-detected infants → **high spontaneous resolution**, documented radiographically. This led to the **Children's Oncology Group (COG) study of observation** for infants with **small perinatal adrenal masses and favourable catecholamine ratios**.

Concept — Regression is the **normal developmental fate of fetal neuroblasts**, so a small favourable-biology tumour in early infancy behaves as a persisting embryonic remnant, not as an early stage of the lethal disease of older children.

Pathology

- **Neuroblastoma** → **ganglioneuroblastoma** → **ganglioneuroma**: increasing maturation.
- **Shimada classification**, revised as the **International Neuroblastoma Pathology Classification (INPC)**: **age-linked**; adds prognostic information **independent of age**. First division: **stroma-poor versus stroma-rich**.
 - **Stroma-poor** is split into favourable and unfavourable by **age, degree of maturation and mitotic rate**; **unfavourable stroma-poor: survival below 10%**.
 - **Stroma-rich: nodular, intermixed, well differentiated**; the last two resemble ganglioneuroblastoma or immature ganglioneuroma and **survive better**.
- **Ganglioneuroma** is benign and fully differentiated. **Metastases maturing into ganglioneuroma** favour origin by maturation rather than de novo. It grows very large before causing compression or **intraspinal extension**. **Survival is independent of extent of resection** → operate only for significant symptoms.

Figure 63.1 legend

- (A) **Poorly differentiated neuroblastoma**, minimal-to-moderate neuropil. (B) **Ganglioneuroblastoma: Schwannian stroma over more than 50% of surface area**; lobule of ganglion cells, immature ganglion cells and poorly differentiated neuroblasts, minimal neuropil.

THE NEUROBLASTIC TUMOUR SPECTRUM

	Neuroblastoma	Ganglioneuroblastoma	Ganglioneuroma
Maturation	Least	Intermediate	Complete; benign
Stroma	Stroma-poor	Stroma-rich (>50%)	Stroma-dominant
Patient	Infants, young children	—	Older children
Site	Adrenal commonest	—	Posterior mediastinum, retroperitoneum; rarely adrenal
Surgery	Risk-adapted	Risk-adapted	Only if symptomatic

- Tests the **direction of maturation**, and the trap that the benign end is the one **not** to chase surgically.

Concept — Histology is graded **against age**: the same immaturity is tolerable in an infant and unfavourable in an older child, and Schwannian stroma marks a tumour maturing towards benignity.

Clinical Presentation and Pattern of Spread

- Usually **abdominal pain or a fixed, hard mass**; otherwise metastatic features — **bone or joint pain, periorbital ecchymosis**. Thoracic tumours: cough, dyspnoea; intraspinal extension: **cord compression**.
- **65% abdominal** (adrenal slightly commoner in children than infants); **pelvic, from the organ of Zuckerkandl, 4%** → **urinary retention and constipation**.
- **Metastases in 70% at diagnosis**.

PARANEOPLASTIC SYNDROMES OF NEUROBLASTOMA

Syndrome	Mechanism	Features
Catecholamine excess	Catecholamine release	Phaeochromocytoma-like: paroxysmal hypertension, palpitations, flushing, headache
Vasoactive intestinal peptide (VIP)	Tumour VIP secretion	Watery diarrhoea, hypokalaemia
Opsoclonus myoclonus ataxia syndrome (OMAS)	Anti-tumour antibodies cross-reacting with neural tissue	Myoclonus, opsoclonus , ataxia; half of OMAS patients have neuroblastoma

Diagnosis**Laboratory Evaluation**

- **Urinary vanillylmandelic acid (VMA) and homovanillic acid (HVA) raised in 90–95%;** they fall with therapy → monitor **response and relapse**.
- Anaemia with marrow disease. Marrow staging: **two aspirates + two biopsies** (biopsy adds substantially); immunocytology may eventually replace biopsy.

Imaging

- Plain film: **calcified abdominal or posterior mediastinal mass**.
- **Computed tomography (CT) and magnetic resonance imaging (MRI)** define local extent and vessels; renal parenchymal invasion is **uncommon** but CT-detectable. **MRI is better** for **intraspinal extension** and great-vessel relationships.
- **Intratumoral calcification and vascular encasement** favour neuroblastoma over Wilms tumour.
- **¹²³I-meta-iodobenzylguanidine (MIBG) scan is required by COG for staging**, replacing skeletal survey and bone scan. Uptake is by **adrenergic secretory vesicles**; it is **more sensitive than ¹³¹I-MIBG or bone scan for relapse**. **Non-avid tumours → positron emission tomography (PET)**.
- Bone metastases: **long bones and skull**.

Screening

- Urinary catecholamine screening in Japan (**>20 years**) aimed to detect earlier stage. Diagnoses **before 1 year rose steeply**, mostly low stage, with almost uniform survival — but **advanced disease after 1 year did not fall**. The aggressive tumours of older children therefore **do not arise from the infant tumours**.
- Screen-detected tumours: **MYCN amplification essentially absent, favourable (hyperdiploid) ploidy**, excellent survival → many would **regress untreated**.

NEUROBLASTOMA SCREENING PROGRAMMES

Programme	Age screened	Detection	Advanced disease / mortality
Japan	Infancy	More infant, low-stage, favourable tumours	No fall after 1 year
Quebec	Infancy	More low-stage tumours	No fall in older-child incidence or mortality
Germany	10–19 months	Fewer early, more unfavourable	No fall in mortality

- At any age, screening **adds favourable tumours without removing unfavourable ones** — the classic example of **overdiagnosis**.

Concept — Screening failed because it detects the **wrong disease**: favourable infant tumours destined to regress, leaving unfavourable disease in older children untouched.

Staging

- Stage determines adjuvant therapy. The **International Neuroblastoma Staging System (INSS)** is **clinical, radiographic and surgical**; systems diverge most in intermediate disease.
- The **International Neuroblastoma Risk Group (INRG) Staging System** is **imaging-based and pre-treatment**: **L1** has no image-defined risk factor (IDRF); **L2** has one or more of **20 IDRFs** (chiefly **vessel or nerve encasement**); **M**; **MS**. **L1** has better 5-year event-free survival (EFS) than **L2**, about **90% versus 78%**.
- The IDRF–resectability link is **disputed**, but IDRFs predict **worse survival and more operative complications** in localised disease. In unresectable disease, IDRFs are **unchanged by chemotherapy in half** and **new in one-fifth**. **Midline perivascular IDRFs** carry more complications and lower EFS and overall survival (OS).
- The **INRG Pretreatment Classification** combines stage, age, histology, grade, MYCN, 11q and ploidy (COG revised version, 2021). It is the **most accurate guide to treatment intensity**.
- Resection-based staging is **institution-dependent**.

TABLE 63.1 INTERNATIONAL NEUROBLASTOMA STAGING SYSTEM (BRODEUR ET AL., 1993)

Stage	Definition
1	Localised, complete gross excision ± microscopic residual; ipsilateral nodes negative (adherent nodes removed with the primary may be positive)
2A	Localised, incomplete gross excision ; non-adherent ipsilateral nodes negative
2B	Localised, non-adherent ipsilateral nodes positive ; enlarged contralateral nodes negative
3	Unresectable unilateral tumour infiltrating across the midline ; or contralateral regional nodes ; or midline tumour with bilateral extension
4	Dissemination to distant nodes, bone, marrow, liver, skin or other organs
4S	Primary as 1, 2A or 2B; spread only to skin, liver and marrow (<10%) ; infants under 1 year

- **Midline = vertebral column**; crossing means infiltrating **to or beyond the opposite side**. The table turns on resection and nodes (1 → 2A → 2B) and on **bone, which excludes 4S**.

INSS VERSUS INRG STAGING SYSTEM

	INSS	INRG Staging System
Basis	Surgical + clinical + radiographic	Imaging, before treatment
Localised	1–3 by resection and nodes	L1 / L2 by IDRFs
Special metastatic	4S, under 12 months	MS, under 18 months
Weakness	Surgeon-dependent	IDRF–resectability link disputed

TABLE 63.2 INRG CLASSIFICATION SYSTEM (COHN ET AL., 2009) — REBUILT BY STAGE

Stage	Qualifiers	Risk
L1 or L2	Ganglioneuroma maturing; ganglioneuroblastoma intermixed	A • Very low
L1, any other histology		
L1	MYCN not amplified / amplified	B • Very low / K • High
L2, any other histology		
<18 months	Not amplified: no 11q aberration / 11q aberration	D • Low / G • Intermediate
≥18 months	Ganglioneuroblastoma nodular or neuroblastoma, differentiating , not amplified: no 11q / 11q	E • Low / H • Intermediate
≥18 months	Poorly differentiated or undifferentiated , not amplified	H • Intermediate
≥18 months	MYCN amplified	N • High

Stage	Qualifiers	Risk
M		
<18 months	Not amplified, hyperdiploid	F · Low
<12 months / 12 to <18 months	Not amplified, diploid	I · Intermediate / J · Intermediate
<18 months	MYCN amplified	O · High
≥18 months	Any	P · High
MS (<18 months)		
MS	Not amplified: no 11q / 11q aberration; amplified	C · Very low / Q · High; R · High

- 5-year EFS: **very low** >85%; **low** >75–≤85%; **intermediate** ≥50–≤75%; **high** <50%. Diploid = DNA index ≤1.0; hyperdiploid >1.0 (includes near-triploid and near-tetraploid).
- Pattern: **MYCN amplification makes any stage high risk** (maturing histologies excepted); **M at ≥18 months is always high**; **11q aberration** lifts non-amplified L2 and MS. Trap: **MS with 11q aberration is high risk**.

Concept — INRG staging moves from *what the surgeon achieved* to *what imaging showed before treatment*, and then lets age and biology, not anatomy alone, set the risk group.

Prognostic Factors

Clinical Variables

- **Age**: historically ≤1 year does better, reflecting **favourable biology**. Age behaves as a **continuous variable**: a cut-off of **460 days** maximised the difference, and INRG places the risk step at **18 months**. **Teenagers and adults** have **indolent but relentless** disease.
- **Site**: **non-adrenal primaries do better**. **Thoracic** tumours present younger and localised and do better **even after correction for age and stage**, being less often amplified and more often **DNA index >1.0**.
- **Stage is powerful and independent**:
 - Stage I completely resected — **all survive**.
 - Stage II is favourable **even when excision is incomplete**.
 - Stages III and IV fare worse.
 - Historical OS for stages I, II and 4S was **75–90%**, against a stage IV 2-year disease-free survival of **19–30%** despite marrow transplantation.
 - Stage distribution is **age-dependent**, and **infants outperform older children at the same stage**.
- **Stage 4S (INRG MS)**:
 - Small primary with spread to **liver, skin and marrow (<10%)**; **no radiographic bone metastases**.
 - **7–10% of cases**; OS **80–88%**; frequent **spontaneous regression**.
 - Adverse features:
 - **neuron-specific enolase (NSE) >100 ng/mL**
 - **ferritin >280 ng/mL**
 - **urinary dopamine >2500 nmol/mmol creatinine**
 - **MYCN amplification**
 - **1p deletion**
 - Deaths occur in infants **under 2 months** with massive abdominal disease, respiratory compromise or **disseminated intravascular coagulation**.

Biologic Variables

- **Homogeneously staining regions and double minutes** (about one-third) are the cytogenetic signature of amplification; **MYCN maps there**. Amplification is **almost always present at diagnosis**, and its adverse effect is **independent of age and stage**. Not every poor outcome is amplified: **many non-amplified advanced tumours still recur**.
- Flow cytometry: **hyperdiploid favourable; diploid and tetraploid adverse**.
- COG risk assignment: **age, stage, MYCN, histological grade, DNA ploidy**.
- **Telomere length** was the sole parameter identifying a **favourable subgroup within high-risk disease**.

BIOLOGICAL PROGNOSTIC MARKERS IN NEUROBLASTOMA

Marker	Frequency	Direction	Association
MYCN amplification (2p24)	20–25% of primaries; ~40% advanced, 5–10% low stage or 4S	Adverse, independent	Advanced stage; 1p loss
1p36 deletion	25–35% overall; ~70% advanced; 70–80% near-diploid	Adverse, independent	MYCN amplification
Unbalanced 11q LOH	—	Adverse, independent	Rarely with MYCN
17q gain	—	Strongest on multivariate	Translocation with 1 or 11
Hyperdiploidy	About half	Favourable	Infants, screen-detected, thoracic
High-affinity nerve growth factor receptor (TRKA) and low-affinity receptor	—	Favourable	Inverse to MYCN
Ferritin, NSE, lactate dehydrogenase (LDH) raised; CD44 absent	—	Adverse, not independent	—

- Trap: in neuroblastoma **diploid is bad and hyperdiploid good**.

Concept — Every marker sorts along one axis — **MYCN amplification and segmental change versus whole-chromosome gain with neurotrophin-receptor expression** — and age is powerful chiefly as a proxy for which side a tumour lies on.

Treatment

- Surgery, chemotherapy and radiotherapy, weighted by **stage, age and biology** (Table 63.2).

Surgery

- Goals: **diagnose → stage → excise if localised → tissue for biology**. Resectability weighs **site, mobility, IDRFs and overall prognosis**.
- **Neoadjuvant chemotherapy improves resectability; never sacrifice vital structures at diagnosis**, especially in young children with excellent prognosis.

Low-Risk Disease (Stages I, II and 4S)

- **Stage I: surgery alone, disease-free survival >90%**. Chemotherapy only at recurrence, **unless MYCN amplification with unfavourable histology; no radiotherapy**. Risk grading now identifies the few who used to recur.
- Resection alone for stages I–II gave excellent OS; supplemental therapy was rarely needed. OS was lower with **MYCN amplification, unfavourable histology, age >2 years and positive nodes**.

- **COG P9641** treated asymptomatic disease with **surgery alone**. Chemotherapy was reserved for symptoms, those at risk of symptoms, or **<50% resection with progression**. Results: 5-year EFS about 89%, **OS about 97%**. In stage 2B, outcome was worse with **unfavourable histology or diploidy** and at **≥18 months**. About one in nine observed patients recurred or progressed.
- **Removing normal organs, particularly the kidney, is not justified**. Radiotherapy to residual stage II disease **does not change progression-free survival** → keep it for chemotherapy-refractory relapse.
- **Stage III, or stage II around the renal vessels: preoperative chemotherapy reduces nephrectomy** (Figure 63.5).
- **4S**: most tumours have **entirely favourable markers**; a small adverse-marker fraction progresses fatally.
 - **Resection of the primary is not mandatory** — survival is equal after complete resection, partial resection or biopsy.
 - Survival falls sharply with **diploidy, MYCN amplification or unfavourable histology**.
 - **Extensive metastases with amplification = high risk** → multimodal therapy. **Favourable and asymptomatic** → supportive care ± limited chemotherapy.

High-Risk and Advanced Disease

- **More than half of new patients are high risk**; the value of extensive resection in stage IV is **contested** (table).
- As systemic control improves, **local control matters more**. Gross total resection plus external-beam radiotherapy gives **84–90% local control**. Double autologous bone marrow transplantation (ABMT) with aggressive surgery and radiotherapy gives **local recurrence of only 2.6%**.
- **Defer resection until after initial chemotherapy**: tumours become **smaller and firmer**, with **less rupture, haemorrhage and nephrectomy**, and **more complete resections**. Timing: **13–18 weeks, after 3–4 courses**; some remain inoperable.
- Complications:
 - **Diarrhoea** from division of **autonomic nerves anterior to the aorta at the coeliac and superior mesenteric origins**.
 - **Chylous ascites** from nodal resection — usually self-limiting.
- **Infants with extensive local or stage IV disease** do better than older children but worse than 4S; **biology separates high from intermediate risk**. In European trials, **non-amplified** infants with midline-crossing primaries or a positive skeletal survey did well **without chemotherapy**; those with overt skeletal, lung or central nervous system metastases did well with **≥4 courses**.

EXTENT OF RESECTION IN STAGE IV NEUROBLASTOMA

Study	Population / comparison	Verdict
Retrospective review	Stages III–IV, radical versus conventional resection	No survival difference
Retrospective review	Stage IV, extent of resection	No survival impact
German NB97 trial	Stage IV, >18 months ; upfront and post-neoadjuvant surgery	No benefit in EFS, local progression-free survival or OS

Study	Population / comparison	Verdict
Two-institution uniform-protocol series	Aggressive surgery + local radiotherapy	No isolated local relapse after >90% resection

- **Survival** endpoints show no benefit; **local-control** endpoints favour aggressive resection — the dispute is partly about **which endpoint**.

Concept — *The value of surgery runs **inversely with biology**: favourable tumours are cured by little or no resection, unfavourable ones are not rescued by radical resection, so vital structures are never traded for margin.*

Chemotherapy

- High-risk sequence: **intensive induction → myeloablative consolidation with stem-cell support → biological therapy for minimal residual disease**. Five-year survival rose from **<15% to 30–40%**; **relapse remains the problem**; dose intensity correlates with response and survival.
- The **Memorial Sloan Kettering (MSKCC) induction** — cyclophosphamide, vincristine and doxorubicin alternating with cisplatin and etoposide — gives high response rates and is widely adopted.
- **Myelosuppression limits dose → ABMT after myeloablative chemotherapy or total-body irradiation**. Complete remission is seen in up to half of stage IV, but with **late relapse**.
- **Bulky disease fails → debulk before ABMT**. Transplant toxicity can be lethal and late effects are unknown; this is **accepted because cure is rare without it**.

HIGH-RISK NEUROBLASTOMA — CONSOLIDATION AND BIOLOGICAL THERAPY TRIALS

Trial	Question	Verdict
German randomised trial	Myeloablation + stem-cell rescue versus maintenance	Better 3-year EFS ; OS not different at first report
COG randomised trial	Myeloablation + ABMT; then 13- <i>cis</i> -retinoic acid versus none	Better EFS; 5-year OS improved on update; retinoid reduced relapse
COG purging trial	Purged versus unpurged marrow	No difference ; stopped early
COG tandem trial	Tandem versus single myeloablation	Tandem improves 3-year EFS
COG anti-GD2 trial	Anti-GD2 + granulocyte-macrophage colony-stimulating factor (GM-CSF) + interleukin-2 (IL-2) + retinoid versus retinoid alone	Antibody superior
Relapse trial	Irinotecan + temozolomide with anti-GD2 versus temsirolimus	Anti-GD2 effective ; mTOR inhibitor very low response

New Innovative Biologic Therapies

- **Double ABMT has reached the ceiling** of myeloablation. **13-*cis*-retinoic acid** differentiates tumour cells, including recurrent ones; it is given **for 6 months after cytotoxic therapy**.
- **GD2 disialoganglioside is uniformly expressed** on neuroblastoma. Antibody is now combined with an **irinotecan–temozolomide backbone**. **Humanised antibodies** aim to reduce the dose-limiting **pain**, and **front-line GD2** trials are planned. Temozolomide and irinotecan (phase I–II in relapse) will advance to front-line trials.
- **¹³¹I-MIBG therapy** delivers radiation via tracer uptake. It achieves responses even after heavy pretreatment (**about 28–39%**) and is now in COG trials.

AGENTS IN NEUROBLASTOMA THERAPY

Agent	Class / target	Role
	Alkylator; vinca alkaloid; anthracycline	

Agent	Class / target	Role
Cyclophosphamide; vincristine; doxorubicin		Induction (MSKCC), alternating with ↓
Cisplatin; etoposide	Platinum; topoisomerase II inhibitor	Induction, alternating cycles
Irinotecan + temozolomide	Topoisomerase I inhibitor + alkylator	Relapse; antibody backbone
13-cis-retinoic acid	Differentiating agent	Maintenance, 6 months
Anti-GD2 (+ GM-CSF, IL-2)	GD2 disialoganglioside	Minimal residual disease after transplant; relapse
Temsirolimus	Mammalian target of rapamycin (mTOR) inhibitor	Relapse — very low response
¹³¹ I-MIBG	Radionuclide via adrenergic vesicles	Relapsed / refractory

- Trap: ¹²³I-MIBG is diagnostic; ¹³¹I-MIBG is therapeutic.

Concept — High-risk therapy is a **staircase of intensification** — induction, myeloablation, tandem transplant, then differentiation and anti-GD2 immunotherapy — each step aimed at the residual disease the previous one left.

Radiotherapy

- **Effective for local control**; relapse risk tracks biology. **No benefit in low stage; improves local control in stage IV and bulky stage III.** No randomised comparison with surgery. Dose **15–30 Gy** by age, site and residual disease.

Spinal Cord Compression

- **Symptomatic in up to 5%; radiographic intraspinal extension in up to 13%.**
- Laminectomy, radiotherapy and chemotherapy give **similar neurological outcomes; severe motor deficits recover little.**
- Sequence: **chemotherapy first → laminectomy only for progressive deterioration** (laminectomy causes **late scoliosis**); **radiotherapy avoided** (impairs spinal growth).

Concept — With equal neurological results from every modality, the choice rests on **late morbidity** — scoliosis after laminectomy, growth arrest after radiotherapy — which leaves chemotherapy first.

KEY POINTS: NEUROBLASTOMA

- **Most common extracranial solid neoplasm** in children.
- Children **younger than 18 months** survive better, attributed to **favourable biology.**
- **Complete resection is curative for low-stage disease.**
- **MYCN amplification** (about 20% of primaries) is associated with **progression** and a poor prognosis **independent of age and stage**; biological markers now drive treatment protocols.

WILMS TUMOUR

Epidemiology

- Wilms tumour (WT), nephroblastoma: **6–7% of childhood cancers, 95% of kidney cancers under 15 years**; an **embryonal tumour** from immature renal remnants.
- Incidence **7.5 per million person-years** (0–14 years). **Over 80% before 5 years**; median 3.5 years; occasionally adults.
- **Younger in bilateral and syndromic disease**, and in males; **slight female excess** overall.

- **Lower in East Asian, higher in Black populations.** The variation follows **race, not geography**, so environmental factors play a minor role. Few exposures are established.
- Black children present **more advanced**; whether biology or **social determinants of health** explain this is unclear. In Africa, survival is poor but improving; **delayed presentation and treatment abandonment** persist.

Biology/Genetics

- **Mostly somatic**; several events are required, so the **Knudson two-hit model does not explain most cases**.
- **10–15%** are due to a **germline variant or early embryonic epigenetic change** (11p15), with or without a syndrome.
- **Familial 1–2%**, two-thirds unexplained.
- Commonest constitutional changes: **WT1 and 11p15.5**. Systematic testing finds a cause in **up to one-third**.
- Whether to refer all patients for genetics is debated. Accepted indications are:
 - **bilateral or multifocal** tumours
 - **age under 2 years**
 - **nephrogenic rests**
 - **overgrowth or predisposition features**, or family history
- In unselected cohorts, **11p15 / Beckwith-Wiedemann spectrum is commonest**, then WT1 syndromes, then **DIS3L2**.

TABLE 63.3 GENE MUTATIONS IN WILMS TUMOURS

Gene (chromosome)	Frequency	Somatic / germline
IGF2 (11p15)	70%	Both
WTX (Xq11)	20%	Somatic only
WT1 (11p13)	20%	Both
CTNNB1 (3p21)	15%	Somatic only
TP53 (17p13)	4% overall; anaplastic ~70% (table) / ~50% (text)	Both

- **11p15 dominates; WTX and CTNNB1 are purely somatic; TP53 is the anaplasia gene.** The two anaplastic TP53 figures come from different sources; "majority of anaplastic tumours" is the safe statement.

WT1

- **First WT gene**, found through **11p13 deletions in WAGR** (Wilms tumour, aniridia, genitourinary anomalies, intellectual disability) and cloned in 1990.
- **WT1 + CTNNB1 (β-catenin) = "ideal type I" WT**; WAGR and **Denys-Drash syndrome (DDS)** tumours belong here (table).
- **DDS = 46,XY disorder of sex development + mesangial sclerosis + WT.**
 - Caused by **zinc-finger point mutations**, over 90% heterozygous germline.
 - The phenotype is worse than deletion because the mutant makes a **dysfunctional (dominant-negative) protein**.
 - **Early nephropathy → end-stage renal disease (ESRD)** in most.
 - Incomplete forms occur, and it has been reported in phenotypic females.
- WAGR and DDS are **more often bilateral and younger**. Renal failure risk is shared by WAGR, DDS and WT1-related **genitourinary anomalies** (rising if they survive to puberty).
- Genitourinary anomalies occur in **4.5% of WT**; WT1 is mutated in only **~2% without them**. The non-tumour kidney has **smaller glomeruli**.

- Only ~20% carry a **WT1 mutation**, so it is not the classic suppressor once assumed. **Somatic WT1: median onset 14 months**, suggesting faster progression.
- **11p15 epigenetic WT = "ideal type II"** (table).
- **Beckwith-Wiedemann syndrome (BWS) and hemihypertrophy:**
 - WT risk 4–10%, **21% bilateral**; age at diagnosis is similar to the general population.
 - Also **adrenocortical tumours and hepatoblastoma**.
 - **Nephromegaly (≥95th centile)** carries the highest risk; genotype now refines risk.

IDEAL TYPE I VERSUS TYPE II WILMS TUMOUR

	Type I (WT1 ± CTNNB1)	Type II (11p15 epigenetic)
Histology	Stromal-predominant	Blastemal or epithelial ; limited nephrogenic differentiation
Rests	Intralobar	Perilobar
Age	Early	Later; heavier birth weight
Associations	WAGR, DDS; male genitourinary anomalies	BWS

WAGR VERSUS DENYS-DRASH SYNDROME

	WAGR	Denys-Drash
WT1 lesion	Deletion (one allele lost)	Point mutation (dominant negative)
Genitourinary	Milder	46,XY DSD
Kidney	Renal failure risk into puberty	Early mesangial sclerosis → ESRD

Familial Wilms Tumor

- **Earlier onset, more bilateral**. Loci **FWT1 (17q12–q21)** and **FWT2 (19q13.4)**, moderate penetrance, **no LOH pattern**.
- Exome sequencing adds **TRIM28** (commonest; **always maternally inherited; epithelial subtype**), **FBXW7**, **NYNRIN** and **KDM3B**.

Other Chromosomal Abnormalities

11p15

- **Imprinted gene cluster; LOH or loss of imprinting in up to 70%**. A WT2 locus is postulated here but its gene is unidentified. The region causes **BWS by epigenetic mechanisms**; Perlman, Sotos and Simpson-Golabi-Behmel act otherwise.
- BWS is mostly sporadic; **up to 15% heritable** (autosomal dominant). **Imprinting centre 1 (ICR1) → IGF2, H19; ICR2 → CDKN1C, KCNQ1OT1, KCNQ1**; parent-specific methylation → **monoallelic expression**.
- Other losses (frequencies in the table): **1p and 16q LOH** → relapse and death. **11q LOH** matters only with **whole-arm loss**.
- **1q gain is not independent of 1p/16q LOH**, since the same translocations produce both. After stage adjustment it still predicts **recurrence and worse OS**.

CHROMOSOMAL MARKERS IN FAVOURABLE-HISTOLOGY WILMS TUMOUR

Marker	Frequency	Significance	Use
16q LOH	~20%	Relapse, death	—
1p LOH	~10%	Relapse, death	—
Combined 1p + 16q LOH	~5%	Adverse at all stages	Regimen M (AREN0532/0533)

Marker	Frequency	Significance	Use
11q LOH	~20%; 3–4× in anaplasia	Adverse only with whole-arm loss	—
1q gain	25–30%	Recurrence, worse OS	Next stratifier

- Combined LOH is the validated stratifier but rare; **1q gain is common**, which is why it is the next target.

Screening

- **Serial renal ultrasonography (US) every 3–4 months** in high-risk children. The interval **does not lengthen with age**, because tumour growth rate is assumed constant.
- Screening detects **lower-stage** tumours but gives **no proven survival benefit**. The gain is **nephron-sparing surgery** in children at risk of bilateral disease (Figure 63.6).
- **Screen if WT risk >1%** (American Association for Cancer Research); some use **>5%**. Continue **until 7 years**.
- **BWS / isolated hemihypertrophy, trisomy 18, Simpson-Golabi-Behmel syndrome**: add **abdominal US + α -fetoprotein** every 3 months from birth to the fourth birthday (hepatoblastoma).
- **Contralateral kidney after nephrectomy**: US every 3–4 months until 5 years. Risk is highest in **infants under 12 months with rests**.
- Suspicious lesion → CT or MRI. **Benign cysts are common in BWS**; recognise them to **avoid unnecessary nephrectomy**.

TABLE 63.4 SYNDROMES ASSOCIATED WITH DEVELOPMENT OF WILMS TUMOUR

Syndrome	Gene	Locus	WT risk
<i>High risk</i>			
WAGR	WT1	11p13	50%
Denys-Drash	WT1	11p13	50%
Fanconi anaemia D1	BRCA2	13q12.3	>20%
Perlman	DIS3L2	2q37	>20%
Mosaic variegated aneuploidy	BUB1B	15q15	>20%
<i>Moderate risk</i>			
Beckwith-Wiedemann	IGF2, H19	11p15	5–10%
Hemihypertrophy	IGF2, H19	11p15	5%
Frasier	WT1	11p13	5–10%
Simpson-Golabi-Behmel	GPC3	Xq26	10% (males)
<i>Low risk</i>			
Li-Fraumeni (TP53, 17p13); neurofibromatosis (NF1, 17q11); Sotos (NSD1, 5q35); trisomy 18; Bloom (BLM, 15q26)			

- The table decides **who is screened** — every high- and moderate-risk syndrome clears 1%. Trap: **BWS is the commonest predisposing syndrome but only moderate risk**; WAGR and DDS are rarer and highest.

Concept — Screening intensity follows risk: the >1% threshold selects the child, the 3-monthly interval reflects growth rate, and the pay-off is **nephron preservation, not survival**.

Pathology

Favorable-Histology Wilms Tumor

- A **pseudocapsule** of compressed parenchyma distinguishes WT from rests. Tumours are **soft, friable, necrotic or haemorrhagic**, which raises the risk of **intraoperative spill**.

- **12% multicentric unilateral.** Extrarenal WT is rare, arising from displaced metanephric or mesonephric remnants.
- Derived from **metanephric blastema**, WT may contain **heterotopic muscle, cartilage or squamous epithelium**. **Classic triphasic pattern = blastema + epithelium** (tubules, rosettes, glomeruloid bodies) + **stroma** (Figure 63.7). Biphasic or monomorphous tumours are diagnostically hard.
- **Epithelial-predominant** tumours have **low aggressiveness and are mostly stage I**, but are **resistant when advanced**.

Anaplastic Wilms Tumor

- Recognised through **National Wilms Tumor Study Group (NWTSG) central review**. **Three criteria**:
 - **nuclear enlargement $\geq 3\times$ adjacent nuclei**
 - **hyperchromasia**
 - **abnormal mitoses**
- Seen in **~2% under 2 years** and **~13% over 5 years**. Causes **chemoresistance**, with a **poor prognosis even at stage I**. **Focal versus diffuse**: diffuse (throughout the kidney or extrarenal) is worse.
- Later age and **absence from rests** mean anaplasia is **acquired** by WT cells; **TP53** is mutated in the majority.

Pathology After Preoperative Chemotherapy

- The International Society of Paediatric Oncology (SIOP) grades response by **volume and histology**.
- **Stromal- and epithelial-predominant tumours are enriched** after chemotherapy: they shrink little but are **excellent if excised**.
- Blastema **responds**, but **residual blastemal predominance $\rightarrow$ relapse and worse survival**.
- Progression during chemotherapy occurs in **~5%** (SIOP 93-01), with lower OS.
- Chemotherapy and radiation may induce **cytodifferentiation** (e.g. rhabdomyomatous), which **confounds biopsy**; differentiated tumours are chemoresistant $\rightarrow$ **excise**.
- Necrosis correlates **positively with pre-treatment blastema** and **negatively with epithelium**.
- **Stage I low-risk after nephrectomy $\rightarrow$ no further chemotherapy**.

SIOP POST-CHEMOTHERAPY HISTOLOGICAL RISK GROUPS

Risk	Histology
Low	Completely necrotic
Intermediate	All others
High	Diffuse anaplasia; blastemal-predominant

Concept — After chemotherapy, **volume response and prognosis come apart**: blastema shrinks but predicts relapse if it survives, whereas differentiated stroma and epithelium barely shrink yet are cured by excision.

Nephrogenic Rests

- Nephrogenic rests (NRs) are found in **more than one-third of WT kidneys** and **over 90% of bilateral cases**. Most **never form WT**: they mature, sclerose, involute or disappear. They are present in **1% of infant kidneys at autopsy**.
- **Methylation** may identify which rests progress.
- **Multiple rests in one kidney imply rests in the other**.
- **Infants under 12 months with WT + rests (especially perilobar)** are at **high contralateral risk** $\rightarrow$ frequent surveillance for years. Those over 12 months with rests are also followed. **Metachronous WT after chemotherapy** shows that rests are not always eradicated.

- **Nephroblastomatosis** = multiple rests. **Diffuse hyperplastic perilobar nephroblastomatosis (DHPLN)**:
 - a thick rind that **enlarges the kidney but preserves its shape** (Figure 63.8)
 - **almost always bilateral**
 - a stand-alone subtype
- WT develops in nearly half with DHPLN, **even after adjuvant therapy**, and **more often anaplastic** after chemotherapy. With 12 weeks of vincristine and dactinomycin, most still developed WT but **all survived** and most renal units kept function.

PERILOBAR VERSUS INTRALOBAR NEPHROGENIC RESTS

	Perilobar (PLNR)	Intralobar (ILNR)
Location	Lobar periphery , subcortical	Anywhere in the lobe , sinus, pelvicalyceal wall
Timing	Late embryogenesis	Early embryogenesis
Appearance	Sharply demarcated ; blastema and tubules	Stroma-rich , intermingled
Syndrome	BWS (11p15)	Aniridia, WAGR, DDS (WT1)
Resulting WT	Type II	Younger ; stromal with rhabdomyogenesis

Concept — Rest position encodes **gestational timing**, and timing encodes **genotype**: early intralobar rests go with WT1, late perilobar rests with 11p15.

Preoperative Evaluation and Staging

- **Over 85% symptomatic** — **mass, pain, haematuria**. Asymptomatic masses are found by parents or doctors, and are **more often stage I–II**. The mass may be huge and cross the midline.
- **Haematuria ~20%; hypertension ~25%** (renin); fever, anorexia and weight loss **10%**; rarely an **acute abdomen from rupture**. Gross haematuria → **cystoscopy + retrograde pyelogram at nephrectomy**, to detect collecting-system extension.
- **Supine-persistent varicocele or hepatomegaly** → **inferior vena cava (IVC) thrombus**; **atrial thrombus** → **hypertension or heart failure** (<10% symptomatic). Examine for **aniridia, hemihypertrophy and genitourinary anomalies**.
- **Operate as an emergency only for active bleeding or rupture**.
- Bloods: **blood count, liver enzymes, electrolytes including urea, creatinine and calcium**.
- **Acquired von Willebrand disease occurs in up to 8%** — check coagulation; **prothrombin and partial thromboplastin times may be normal**. Correct with **desmopressin (DDAVP)**, or intravenous immunoglobulin or plasmapheresis if needed; it **reverses after resection or chemotherapy**.

Imaging

- Paediatric renal tumours **overlap radiologically**: WT cannot be reliably separated from renal cell carcinoma (RCC) or clear cell sarcoma of the kidney (CCSK). A minority of presumed WTs treated without biopsy prove to be **other diagnoses** (SIOP-9; United Kingdom Children's Cancer Study Group, UKCCSG).
- Clinical clues:
 - A syndrome → WT; **bilateral or multicentric** → WT (lymphoma can mimic).
 - **Neonate** → **congenital mesoblastic nephroma (CMN)**, though favourable-histology WT and **rhabdoid tumour of the kidney (RTK)** also occur early.
 - WT can mimic neuroblastoma.
- In National Wilms Tumor Study (NWTs)-3, misdiagnoses were mostly **unimaged** children, who had **more complications**. What matters is establishing a **solid renal tumour**, so that a **major cancer operation** is planned.

- Sequence: **US first → contrast CT or MRI of abdomen and pelvis in all → chest CT**. US triages; avoid non-contrast CT.
- Doppler can suggest caval extension (~4%), but **phased CT is more reliable**. SIOP favours MRI with **diffusion-weighted imaging**; COG accepts either. MRI characterises better but **neither differentiates nor stages reliably**, and needs anaesthesia.
- Cross-sectional imaging shows extrarenal spread and synchronous lesions (Figure 63.9). **No contralateral exploration if imaging is normal**.
- **Rupture**: CT has **poor sensitivity**; **ascites beyond the cul-de-sac** is the best predictor.
- **Nodes and extrarenal invasion**: imaging is unreliable, and **positron emission tomography (PET)** adds **nothing** → **inoperability is decided at exploration**.
- **Lung = commonest metastatic site** → **chest CT**, done before MRI to avoid **anaesthetic atelectasis**.
- **CT-only nodules** are controversial. Some treat with dactinomycin (actinomycin D, AMD) and vincristine (VCR) alone; others report more relapse without intensification.
- **AREN0533**: no 1p/16q LOH + **complete nodule response at week 6** of VCR, AMD and doxorubicin (DOX) → **lung irradiation omitted**, with no loss of EFS or OS against NWTS-5.
- Surveillance:
 - **COG**: chest CT with abdominal CT/MRI alternating with US.
 - **SIOP**: US + chest radiograph only.
 - In NWTS-5, symptom-detected relapse did worse, but **imaging modality made no difference**.
- **CCSK and RTK** → **bone scan + cranial CT/MRI**.

Staging

- **Histology and stage are the key determinants**; COG staging is **surgical-pathological**.
- **Renal sinus vessel invasion is the first sign of spread (stage II)**, then capsular penetration, which requires **tumour in perirenal fat**. **Any spill** → **stage III**.
- **Stage III is commonest** in favourable histology; anaplastic tumours present more often at III–IV.

TABLE 63.5 STAGING SYSTEM OF THE CHILDREN'S ONCOLOGY GROUP

Stage	Description
I	Confined, completely resected; capsule intact; no rupture ; no renal sinus extension
II	Extracapsular but completely resected ; renal sinus extension or extrarenal vessel thrombus/infiltration
III	Residual abdominal tumour: nodes, any spill, peritoneal implants, positive margin, incomplete removal
IV	Haematogenous metastases (lung, liver, bone, brain)
V	Bilateral at diagnosis

- Tests the **II / III boundary**: extension completely removed is II; anything **left or spilled, including nodes**, is III.

Prognostic Factors

- Better therapy has **blunted traditional factors**, so trials now stratify by biology.

Chromosomal Abnormalities

- **16q and/or 1p LOH** predicts relapse **independent of histology and stage**. In NWTS-5, stage I–II was worse with **either** LOH; stage III–IV only with **both**.
- **AREN0532/0533**: combined LOH → **regimen M** (cyclophosphamide + etoposide added to VCR, AMD, DOX) → better EFS at all stages. Stage III–IV EFS was **~90% versus 61%** on NWTS-5.
- Immunohistochemical predictors lie in the **blastemal component**; they need validation.

AGENTS IN WILMS TUMOUR THERAPY

Agent	Class	Use
VCR	Vinca alkaloid	Backbone; SIOP two-drug preoperative
AMD	Antitumour antibiotic	Backbone; SIOP two-drug preoperative
DOX	Anthracycline	Three-drug regimens; bilateral (COG); cardiotoxic
Cyclophosphamide + etoposide	Alkylator + topoisomerase II inhibitor	Regimen M; CCSK
Carboplatin	Platinum	Anaplastic, high-risk, relapsed; gonadotoxic

Concept — Wilms stage is **surgical-pathological** — spill, nodes, margins — so the operating surgeon, not the scan, sets chemotherapy intensity and whether the child is irradiated.

Treatment**Surgical Considerations**

- **Radical nephrectomy, transperitoneal.**
- Explore for **local extension, liver and nodal metastases, peritoneal seeding.**
- **Palpate renal vein and IVC before ligation**, using intraoperative US when equivocal. IVC extension occurs in ~6%, and more than half are asymptomatic.
- **Spare the adrenal** if it is clear of tumour.
- **Lymph node (LN) sampling is essential; formal dissection is not.** En bloc resection yields fewer positive nodes.
- **Omitted sampling is the commonest protocol violation (10–20%)** and causes under- and over-staging. **More nodes sampled → more positives found.**
- Basins: **hilar; paracaval (right); para-aortic (left); interaortocaval (both).** Skip metastases occur interaortocaval.
- **Node yield:** 5–10 is ideal, with **at least 5–7** sampled. COG requires ≥ 5 ; ≥ 6 is achievable after SIOP chemotherapy. Extended sampling adds no morbidity.
- LN density predicts survival, but thresholds vary widely (**0.22–0.65**).
- **Avoid spill:** it occurs in ~10%, more with **right-sided and larger** tumours, and causes **local abdominal relapse.**
- **Local recurrence risk factors:** spill, unfavourable histology, incomplete removal, no LN sampling (highest in stage II). With all spill now staged III, the stage III risks are **positive nodes, combined LOH and residual disease**; spill itself no longer predicts relapse, because extra therapy is given.
- **Minimally invasive surgery (MIS):**
 - Usually done **after chemotherapy**; cases are few.
 - **Oncology must not be compromised**, and upfront MIS raises spill concerns.
 - SIOP UMBRELLA accepts it for **central tumours unsuitable for nephron-sparing surgery (NSS).**
 - Node sampling is harder.
- **Complications ~11% (NWT5-4)**, chiefly **haemorrhage and small-bowel obstruction.** Risk factors: **higher stage, >10 cm, wrong diagnosis, thoracoabdominal incision, caval extension, other-organ resection.**

IMMEDIATE VERSUS DELAYED NEPHRECTOMY

Immediate (COG)	Delayed after 6 weeks of chemotherapy (UKCCSG / SIOP)
Unaltered histology and biology (LOH 1p, 16q, 1q gain) for risk assignment; very-low-risk cured by nephrectomy alone (AREN0532)	Fewer complications; spill near zero
Higher spill and complication rates	Loses pre-treatment staging and biology; a minority not WT without biopsy

COG VERSUS SIOP

	COG	SIOP
Unilateral primary	Upfront nephrectomy	Preoperative chemotherapy, no biopsy
Risk basis	Stage + histology + biology	Post-chemotherapy histology
Primary imaging	CT or MRI	MRI + diffusion-weighted imaging
Bilateral	VCR + AMD + DOX	VCR + AMD if non-metastatic
Unilateral NSS	Limited	UMBRELLA criteria
Follow-up	CT-based	US + chest radiograph

Concept — COG and SIOP trade the same two goods — **untreated biology for risk assignment** against a **safer, spill-free operation** — and every procedural difference follows from which is prioritised.

Adult Wilms Tumor

- Once poor, now improved → **stage-appropriate combined-modality therapy**.

Relapse**RELAPSED WILMS TUMOUR RISK GROUPS**

Group	Definition	Salvage survival
Standard	Non-anaplastic, after VCR and/or AMD only	70–80%
High	Non-anaplastic, after ≥3 agents	40–50%
Very high	Anaplastic or blastemal recurrence	~10%

- Survival after relapse **falls with prior treatment intensity**: relapse through three drugs means a resistant tumour.
- The role of high-dose chemotherapy with stem-cell rescue is undetermined.
- AREN03B2 relapses: **SIX1 or MYCN-network mutations in over 40%; 1q gain in 75%**.
- Surgery in relapse is under study (AREN1921).

Bilateral and Bilaterally Predisposed Wilms Tumors

- **Synchronous bilateral WT 5–7% → no initial radical nephrectomy; chemotherapy → renal-sparing surgery.**
- Renal failure **~15% at 15 years** (bilateral); ESRD up to **20% metachronous**. **Commonest cause: bilateral nephrectomy for persistent or recurrent tumour**; avoidable with a careful protocol and experienced surgeons.
- AREN0534 sequence: **VCR + AMD + DOX for 6 weeks, no biopsy → imaging at week 6 → surgery if feasible → otherwise more chemotherapy, but surgery by week 12.**
- Rationale: in NWTS-4, poor responders had prolonged chemotherapy, reached surgery at a **median of 7 months**, and some **became anaplastic**.

- Non-responders → **bilateral open biopsy**. **Open** because it detects anaplasia better than needle biopsy; **bilateral** because anaplasia is **discordant between kidneys in ~4 of 5**.
- **Volume change does not predict histology**; non-shrinkage usually means **differentiation**, which is excellent if excised (Figure 63.11: minimal shrinkage at 12 weeks; **mature rhabdomyoblastic elements** at partial nephrectomy).
- Results: **84% operated by week 12**; **bilateral NSS 39%** (target 50%), the rest mostly NSS on one side; **4-year EFS 82.1%, OS 94.9%**, against **56% and 80.8%** on NWTS-5.
- Some centres achieve **bilateral NSS in over 90%**, because compressed parenchyma is **easily underestimated** (Figure 63.12) → **consider NSS in all**.
- Enucleation gives **positive margins in 23–36%**, with good outcome **if flank radiotherapy is given**.
- In anaplasia (AREN0534), histology was **discordant in 80%**. In diffuse anaplasia, margins did not matter (all irradiated), but **outcome was poor**.
- **Bilateral nephrectomy and dialysis** are needed rarely, for refractory tumour. **Modified chemotherapy remains possible** when anephric.
- **Transplant after 1–2 years**; **remove residual renal tissue first**, to prevent recurrence under immunosuppression.
- Follow-up is **long term**, since **late relapse can occur after 4 years**; monitor renal function, proteinuria and blood pressure.
- **SIOP**: same principle (no biopsy, NSS by week 12) but **two-drug AMD + VCR if non-metastatic**.
- **Predisposition syndromes / multicentric tumours** (AREN0534): over half kept both kidneys; **4-year EFS 94%, OS 100%**.

Partial Nephrectomy for Unilateral Tumors

- **Renal failure after unilateral nephrectomy is only 0.6% at 20 years**. The risk is higher in **genitourinary anomalies, DDS and WAGR**, because WT1 is needed for renal development. **Syndromic children** have small screen-detected tumours suited to NSS.
- Most tumours are too large. After chemotherapy, NSS is possible in **10–15%** with an aggressive approach, **~3%** under the SIOP 2001 criteria (polar, peripheral, non-infiltrating), and **<10%** even in very-low-risk stage I.
- Concerns are **staging after chemotherapy** and **local recurrence**; **intra-abdominal relapse markedly reduces survival**.
- **SIOP 2016 UMBRELLA NSS criteria**: **<300 mL**, substantial remnant, negative nodes. **MIS for NSS** has scant evidence and risks **upstaging by spill**.
- The adult **4-cm small-renal-mass cut-off** helps; manage a small mass by NSS after **multidisciplinary review**, with a **margin of normal kidney** and **frozen section**.

Concept — In bilateral disease the aim shifts from clearance to **nephron preservation within a fixed 12-week clock**; *prolonging chemotherapy for shrinkage buys differentiation and anaplasia, not parenchyma*.

Management of Intravascular Extension

- Thrombus into the renal vein, IVC or atrium occurs in **~10%**.
- Consensus: **thrombus at or above the hepatic veins → preoperative chemotherapy**, to avoid bypass.
- **Viable tumour remains in 36–60%** of thrombi, without affecting survival.
- Chemotherapy **beyond 6–12 weeks adds no regression** and is discouraged.
- **COG**: **upfront nephrectomy below the hepatic veins**, with the thrombus removed **en bloc**. **Piecemeal removal → stage III → radiotherapy**.

- In a multicentre series:
 - **Higher thrombi responded better**, and nearly half regressed in level.
 - **Bypass was avoided in two-thirds** of suprahepatic cases.
 - Complications were commoner with upfront surgery.
- The role of anticoagulation is unclear.

Concept — *Thrombus level sets the sequence because the aim is **avoiding bypass**, not sterilising the thrombus — residual viable cells do not alter survival.*

Late Effects of Treatment

- Childhood cancer **5-year survival exceeds 80%**, at the cost of normal-tissue injury. Data come from the **NWTSG Late Effects Study** and the **Childhood Cancer Survivor Study (CCSS)** cohorts (1970–1986 and 1987–1999).
- WT survivors: **any chronic condition 65% at 25 years**, still rising; **severe (grade 3–4) 24%**.

Mortality

- Risk of death stays **elevated for years**. It **fell for patients treated in the 1990s**, reflecting **less radiotherapy and fewer anthracyclines**.

Fertility and Pregnancy

- **Males:** gonadal radiation causes dose-dependent **hypogonadism and temporary azoospermia**. **Leydig cells are more radioresistant than germ cells**; high doses give low testosterone and **delayed puberty**. Chemotherapy also harms the testis.
- **Females:** **pelvic radiation and alkylators** cause **ovarian failure and premature menopause**; **few pregnancies follow whole-abdominal radiotherapy**.
- Pregnancy after radiotherapy:
 - Offspring are at risk of **low birth weight and prematurity**.
 - **Pelvic fields or doses >20 Gy** raise miscarriage.
 - **Hypertension in pregnancy** is commoner after abdominal radiotherapy.
- Most WT patients are **low gonadotoxic risk**. **Cyclophosphamide, carboplatin and whole-abdomen irradiation** (anaplastic, high-risk and relapsed disease) raise it → **counsel families**. Prepubertal **tissue cryopreservation** is experimental but increasing.

Second Malignancies

- **1% at 10 years**, rising thereafter. **Radiation is a leading risk**, with tumours mostly **in-field**. **Leukaemia peaks in the first 5 years**; solid tumours appear later.

Cardiac Effects

- **Congestive heart failure in 4.4% after DOX**, higher with **whole-lung or left-flank irradiation**. Almost none occurred below the **contemporary cumulative dose of 150 mg/m²**; subclinical toxicity is unassessed. Cardiac disease is the **second commonest cause of death >30 years** after WT (British Childhood Cancer Survivor Study).

Concept — *Late effects track two exposures — **radiotherapy and anthracycline** — so falling late mortality mirrors their reduction, and every trial that safely omits irradiation is also a late-effects trial.*

Other Renal Tumors

Tumour-by-tumour detail follows; the comparison table at the end of the section carries the shared attributes.

Clear Cell Sarcoma of the Kidney

- CCSK is **3%** of renal tumours, named for its clear cytoplasm.
- Survival is better with **lower stage, younger age, DOX and absence of necrosis; adding DOX improved survival.**
- Stage I survival is **98%**, yet **stage I still needs radiotherapy.**
- **Late relapse** (~30% after 3 years; up to 10 years) → long follow-up.
- **No bilateral cases, no WT-associated anomalies.**
- Four-drug NWT5-5 therapy was no better than NWT5-4; **stage is strongly predictive.**
- **BCOR internal tandem duplication** is detectable in cell-free DNA.

Rhabdoid Tumor of the Kidney

- **Most lethal** renal tumour of childhood, **2%**. It occurs renal and extrarenal (as does CCSK), suggesting a **non-organ-specific mesenchymal origin.**
- **Biallelic SMARCB1/INI1 loss (22q11.2)**, germline in one-third; **INI1 immunostaining is negative.**
- Features: **median ~16 months**, advanced stage, chemoresistance, **brain metastases; younger is worse.**
- There is **no standard therapy**, and high-dose chemotherapy with stem-cell transplant failed.
- **EZH2 is upregulated** → **tazemetostat**; a COG trial is planned.

Congenital Mesoblastic Nephroma

- **Commonest infant renal tumour** (mean **3½ months**) and **commonest antenatal renal tumour.**
- **Firm, trabeculated, leiomyoma-like.** Subtypes: **classic, cellular, mixed.**
- **ETV6-NTRK3 fusion**, shared with congenital fibrosarcoma. **Translocation-positive cellular tumours relapse less. WT1 is not expressed;** the tumour arises from **stromagenic blastema.**
- **Radical nephrectomy alone usually cures**, but it may extend into the hilum → **complete resection.**
- **Cellular** tumours can recur or metastasise; risk is lower under 3 months.
- **No routine adjuvant therapy;** consider it for **incompletely resected cellular** tumours. Inoperable or recurrent tumours respond to chemotherapy.

CLASSIC VERSUS CELLULAR MESOBLASTIC NEPHROMA

Classic	Cellular
Bland spindle-cell sheets; like infantile fibromatosis; low recurrence	Solid sheets, frequent mitoses; like congenital fibrosarcoma; recurrence and metastasis

Solitary Multilocular Cyst and Cystic Partially Differentiated Nephroblastoma

- **Multilocular cystic nephroma:** benign. Half occur in **young boys**, with a second peak in **young adult women;** rarely bilateral. It is encapsulated, with **only mature cells in the septa.**
- Cured by nephrectomy. If partial nephrectomy is attempted, use **frozen section to exclude cystic partially differentiated nephroblastoma (CPDN) or CCSK.**
- Associated with **DICER1 syndrome** (autosomal dominant, reduced penetrance; also predisposes to WT). The distribution is **bimodal:** two-thirds before 4 years, the rest in the fourth to sixth decades.
- **CPDN:** first 2 years; **radiologically identical; blastema or rests in the septa;** possibly the same entity. **Surgery cures, with 100% survival.**

Metanephric Adenofibroma

- A stromal tumour **resembling CMN;** its epithelium ranges from **metanephric adenoma to WT**, with papillary RCC-like areas.

- Possibly **ILNR-derived**; with a WT component it presents at a **mean of 12 months**.
- **No recurrence** after nephrectomy with WT chemotherapy.

Renal Cell Carcinoma

- **Commonest renal malignancy of the second decade**, though **~5% of all RCC** occurs in children. **Haematuria is commoner than in WT**, and imaging cannot distinguish them.
- **Translocation RCC (~50%) is the commonest histology**:
 - involves **TFE3 (Xp11.2)**, a microphthalmia transcription factor (MiT) family member, or TFEB
 - epithelial markers are weak
 - nodes are involved in **40–50%**, and stage is higher
- Papillary RCC is relatively commoner than in adults.
- **Renal medullary carcinoma: sickle cell haemoglobinopathy**, median 13 years, **mostly stage IV**, lethal; linked to **SMARCB1**.
- **~5% are syndromic**: von Hippel-Lindau, Birt-Hogg-Dubé, tuberous sclerosis, hereditary papillary RCC, hereditary leiomyomatosis and RCC, succinate dehydrogenase RCC. **Young onset → genetic evaluation**.
- Subtype order (AREN03B2): **translocation > papillary > medullary > chromophobe > clear cell > fumarate hydratase- > succinate dehydrogenase-deficient**. Syndromes were found in under one in ten, **tuberous sclerosis commonest**.
- **Complete resection is the key determinant: all stage I survive**; stage I–II **64–80%** in older series. Partial nephrectomy data are limited, and **younger age is favourable**.
- **Nodes are less ominous than in adults** (translocation RCC with positive nodes ~93% survival). **Chemo- and radioresistant**.
- COG prospective study: median age **13 years**, the **13–14-year crossover** above which RCC outnumbered WT. Translocation RCC ~60%; node-positive just over 30%. Resected stage I–II was excellent, and node-positive disease did better than in adults. SIOP results are similar.

Angiomyolipoma

- **Hamartoma**, rare in children. Adults are mostly sporadic; children have **tuberous sclerosis complex (TSC)**, often **bilateral**.
- Angiomyolipoma (AML) occurs in **50–80% of TSC**. It enlarges through childhood, with a **growth spurt in adolescence**.
- Growth is fastest **over 11 years** and in lesions **<2 cm → annual US**, with **MRI** when both apply.
- Other TSC lesions: **cysts (up to 30%)**, polycystic disease, RCC.
- **TSC1 (chromosome 9) / TSC2 (chromosome 16)** mutations are found in 85%; **LOH** drives growth.
- Risks: **haemorrhage and renal impairment**.
- **First-line for growing lesions: mTOR inhibitor** (Figure 63.13); embolisation or surgery is second-line.
- Embolisation infarcts adjacent parenchyma, does not prevent new lesions, and needs **repeating in 35% → reserve it for active bleeding**.
- **TSC loss hyperactivates mTOR**. Sirolimus shrinks lesions **only while taken**, and they **regrow after stopping** → continuous therapy.
- **EXIST-2 (everolimus, adults): ≥50% volume reduction in 42% versus 0% on placebo**; extension showed **no bleeding or nephrectomy**.
- **Fat-poor AML mimics RCC → biopsy**. Use **nephron-sparing approaches in TSC**.

ANGIOMYOLIPOMA INTERVENTION THRESHOLDS

Source	Threshold
Historical	>4 cm
Adult sporadic AML	May wait until >6 cm
International TSC Consensus Group	>3 cm; >3 cm and enlarging = greatest bleeding risk

Concept — TSC angiomyolipoma is a **pathway disease**, so pathway blockade is first-line and continuous, whereas embolisation and surgery treat one lesion in a kidney that will keep producing more.

PAEDIATRIC RENAL TUMOURS — COMPARISON

Tumour	Age / share	Genetics	Spread	Management signature
Wilms tumour	Median 3.5 years; 95%	WT1, 11p15, CTNNB1, WTX	Lung , liver	Stage-, histology-, biology-adapted
CCSK	3%	BCOR ITD	Bone, brain ; late relapse	DOX chemotherapy; radiotherapy even at stage I
RTK	~16 months; 2%	SMARCB1/INI1	Brain	No standard; most lethal
CMN	~3½ months	ETV6-NTRK3	Rare (cellular)	Nephrectomy alone
Cystic nephroma / CPDN	Young children; adult women	DICER1	None	Complete excision
RCC	Second decade	TFE3/TFEB; SMARCB1 (medullary)	Nodes, less ominous	Complete resection
AML	TSC, adolescence	TSC1/TSC2	—	mTOR inhibitor

- Organises the differential by **age** (neonate CMN → toddler WT/RTK → adolescent RCC) and by **metastatic site** (lung WT; bone CCSK; brain RTK/CCSK), which sets the **staging investigations**.

KEY POINTS: RENAL TUMOURS

- **WT1 deletions (11p13)** occur with **aniridia** and **WT**; **WT1 mutations** occur in **DDS**.
- **Local recurrence** rises with **spillage** and **nodal involvement**; survival falls markedly after relapse.
- **Bilateral, multifocal and syndrome-associated unilateral WT** → **preoperative chemotherapy**, allowing renal-sparing surgery to reduce renal failure.
- **CMN is the most common renal tumour in infants**.